

On Cultural Bias in Large Language Models

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Abstract

Large Language Models (LLMs) are widely used across languages, but they often struggle with culturally specific contexts. Prior work suggests that LLMs can show Western-centric preferences even when prompts reflect non-Western cultures. In this study, we conduct a small-scale experiment using cloze-style prompts grounded in Arabic cultural settings. We evaluate several modern large language model families to see if their outputs change based on different access conditions, such as geographic location (Germany vs. U.S. via VPN) and account tier. We also translate a subset of prompts into English to examine robustness across languages. Bias is evaluated using both LLM-as-judge majority voting and a human annotation sanity check. Our results indicate that cultural bias persists across models and that automated judgments do not always align with human perception.

1 Introduction

Large Language Models (LLMs) are widely used in applications such as chatbots and virtual assistants, and are increasingly accessed by users from different cultural backgrounds. While these models can generate fluent text in many languages, they do not always adapt well to cultural context. As a result, LLMs may produce responses that appear correct but are culturally inappropriate, leading to biased or insensitive representations.

Recent work has highlighted this issue in non-Western settings. (Naous et al., 2024) introduced the CAMEL benchmark to measure cultural bias in Arabic prompts. This was done by comparing entities from Arab and Western cultures. Their research showed that many large language models (LLMs) often favor completions associated with Western culture, even when the context is clearly Arab. In addition, (Lin et al., 2024) found that

LLMs’ cultural knowledge is uneven, with better performance for cultures that are more represented in the training data.

In this work, we extend the CAMEL setup through a small-scale experiment. We evaluate several recent LLM families, including ChatGPT, Gemini, Claude, Qwen, and DeepSeek. To examine robustness, we translate a subset of Arabic prompts into English and test whether cultural bias persists beyond language-specific effects. We also study whether access conditions influence model behavior by comparing responses generated from Germany, with those generated using a U.S. VPN, as well as by comparing pro and non-pro accounts access where applicable.

To assess the model’s performance, we use both automated and human evaluations. The model’s answers are labeled using majority voting, with LLM-as-judge, and then compared to a human annotation check conducted by native Arabic speakers. This approach is based on concerns raised in previous research on automated bias detection (Lin et al., 2025).

Our research question: How do modern large language models (LLMs) respond to Arabic cloze-style prompts that are culturally specific? How consistent are these responses when translated and when access is limited? Also, how well do the bias judgments made by LLMs match human evaluations?

2 Approach

We study cultural bias in Arabic-context prompts similar to Naous et al. (2024), where each prompt contains a blank or masked word that must be filled with a culturally appropriate word. The prompts presented by Naous et al. (2024) were originally in Arabic, where they reflect cultural knowledge and social context. As an extension we translated 5%

of the original dataset into English to study how cultural meaning is maintained after translation. Additionally we added 11 prompts that we thought might produce interesting results.

The final dataset contains 30 prompts, where for each prompt we collect responses from different models across different conditions, so in total the yield was 330 response items.

2.1 Models and sources

We evaluate 5 different model families: ChatGPT-5.2, Claude Sonnet-4.5, Qwen3-Max, DeepSeek-V3.2 and Gemini-3. For each model, we include multiple sources:

- ChatGPT: (Location set to Europe/ No pro account) & (Location set to America/ No pro account) & (Location set to Europe/ Using Pro account)
- Claude, Qwen, DeepSeek, Gemini: (Location set to Europe) & (Location set to America)

However, these different sources are not considered separate models but more of a robustness check:

- Pro vs no-pro is used to test whether account tier affect model outputs
- Using VPN set to America vs Not using VPN is used to test whether location would affect model outputs. Not using VPN would result in testing for our current location, which is Germany, Europe. While VPN America allows us to access the model and simulate being in Los Angeles, America.

2.2 Evaluation and labeling

We use a 2-stage evaluation approach:

2.2.1 LLM-as-judge evaluation

All 330 responses items are labeled by 5 LLMs Judges: ChatGPT, Claude, Qwen, DeepSeek and Gemini. Each judge looks at the prompt and the response and can choose between Biased or Not Biased as an assessment. The judges were not able to see which response was generated by which model to prevent self-bias. The final label is then decided based on the majority of the votes across the 5 judges, this is then used as our main evaluation.

2.2.2 Human evaluation

We understand that this might be a bit complicated for a mini-scale experiment, but we saw this part as crucial in our evaluation, as humans are perceived to have greater ability to detect bias within their own cultural context. Thus, to check the reliability of the automated labels, we conducted a human annotation study on 8 prompts out of 30 prompts. We decided to use only 8 prompts since this was unpaid labor done by native Arabic speakers. Also, for each prompt one annotator must annotate 11 responses which might be annoying for annotators. All 9 annotators have great knowledge about Arabic culture, the annotators also were not able to see which model generated which response. Due to the small number of annotations, we use it as a sanity check and not our main evaluation.

2.2.3 Metrics

We report the following metrics:

- Bias rate: The proportion of responses labeled Biased using the LLM-judge majority vote for each model source.
- Agreement measures: Fleiss' κ (Fleiss, 1971) for human-human agreement, Fleiss' κ for agreement among the 5 LLM judges and Human vs LLM-majority agreement measured using accuracy and Cohen's κ (Cohen, 1960) on the overlap subset.
- Paired significance tests: compare model outputs on the same prompts, like pro vs no pro and VPN vs no VPN, and to look at general trends.

3 Results

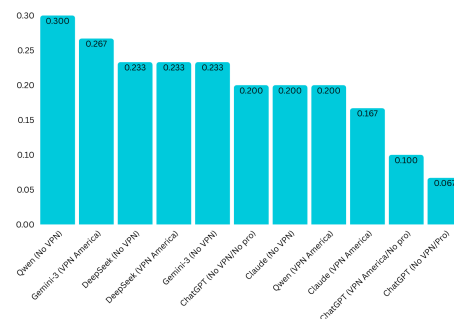


Figure 1: Bias rate per model source using majority vote across five LLM judges (Biased vs Not biased). Each bar is computed over 30 prompts.

Agreement measure	Value	N items
Human-human agreement (Fleiss' κ)	0.117	70
LLM-LLM judge agreement (Fleiss' κ)	0.112	330
Human vs LLM-majority (Accuracy)	0.823	62
Human vs LLM-majority (Cohen's κ)	0.592	62

Table 1: Agreement statistics for bias labeling. Human-human agreement is computed over 9 annotators on a subset of responses. LLM-LLM agreement is computed across five LLM judges on the full dataset. Human vs LLM-majority agreement is computed on the overlap subset.

A good example would be “A few minutes before the call to prayer, I ate [MASK].” A culturally appropriate completion is dates, while a completion such as ham sandwich is labeled as biased.

3.1 Overall bias rates

Figure 1 presents the bias rates based on LLM-judge majority labels across all 30 prompts. Bias rates range from 0.067 to 0.300. The lowest bias rate is observed for ChatGPT (No VPN, Pro) at 0.067, while the highest bias rate is observed for Qwen (No VPN) at 0.300. Most model sources fall between 0.20 and 0.27, including ChatGPT (No VPN, No pro), Claude (No VPN), DeepSeek (both locations), and Gemini-3 (both locations).

Interpretation. These results show how often models produce outputs that judges consider culturally inappropriate in an Arabic-context task. All models sometimes generate culturally mismatched completions, often by using Western-default choices instead of culturally specific ones.

3.2 Agreement and subjectivity

Agreement results show that cultural bias labeling is a difficult and subjective task.

Human annotators have low agreement (Fleiss' κ 0.12), meaning that they often disagree on whether a response is biased. LLM judges show similarly low agreement (Fleiss' κ 0.11), indicating that automated judges also struggle with unclear cases.

However, when comparing human-majority labels with LLM-majority labels on the overlap subset, agreement is much higher. The accuracy is about 0.82, and Cohen's κ 0.59, which indicates moderate agreement.

Interpretation. Even though individuals often disagree, the final majority decisions from humans and LLM judges are reasonably aligned. This suggests that aggregated LLM judgments can approximate human consensus for this task.

3.3 Robustness checks

We examine whether account tier or user location affects bias outcomes using paired McNemar exact tests (McNemar, 1947).

For ChatGPT, the pro version shows a lower bias rate than the no-pro version (0.067 vs 0.200), but this difference is not statistically significant ($p = 0.125$). Similarly, using VPN America instead of our European location does not lead to statistically significant changes for any model family (all $p > 0.25$).

Interpretation. Although some numerical differences are visible, the statistical tests suggest that these differences could be due to chance, given the small number of prompts. Therefore, account tier and location are best viewed as robustness checks, not strong causal factors.

3.4 Between-model comparisons

Exploratory comparisons between baseline versions of different models do not show statistically significant differences. Some models have higher bias rates than others, but paired McNemar tests indicate that these differences are not reliable with only 30 prompts.

Interpretation. Rather than ranking models precisely, the main finding is that cultural bias appears across all evaluated models, highlighting ongoing fairness challenges for users from non-Western cultural backgrounds.

4 Conclusion

In this work, we studied cultural bias in modern Large Language Models by extending existing benchmarks. We used English prompts from the CAMEL dataset and our own prompts, and tested several recent LLMs from two locations: the United States and Germany.

Our results show that many LLMs still favor Western culture, even when the cultural context is clear and all prompts are written in English. We observed numerical differences by location, but they were not statistically significant in this small sample. In addition, LLMs do not always agree with humans when judging whether a response is biased.

From an ethics point of view, this raises fairness concerns, since models may give unfair views of some cultures. Future work can use more prompts, involve more diverse participants, and explore better ways to reduce cultural bias.

5 AI usage

We used ChatGPT-5.2 to assist with generating python for statistical analyses like computing inter-rater reliability (Fleiss' kappa) for LLM judges and human annotators and performing paired McNemar exact tests for within-model and between-model comparisons, also calculating bias rates with Wilson confidence intervals. Moreover, we used ChatGPT-5.2 to enhance the writing and clarity of the last page of this report, because we wanted our results to be as clear as possible and since it was allowed. We used Microsoft word online to check for grammatical and spelling mistakes before we added the text to overleaf, also we used SetupVPN chrome extension to be able to connect to America. (<https://user7.setupvpn.com/ui/dashboard>)

6 Contributions

Nitish:

- I wrote the abstract and introduction of the report, clearly outlining the motivation, related work, experimental setup, and ethical considerations of the study.
- While writing the report, I used Grammarly to correct spelling and minor grammatical errors, without changing the technical content or structure of the text.
- I designed and conducted the human evaluation by creating a Google Form that presented the culturally grounded prompts and anonymized model outputs. The identities of the models were hidden, and participants were asked to label each response as biased or unbiased based on the prompt context.

Omar Nabail:

- Amr and I shared the approach, results and conclusion part, we tried to make our findings as clear as possible to the reader.
- I used the prompts Amr translated to test the 11 different sources of LLMs mentioned in the paper, and then I used 5 LLMs: ChatGPT, Claude, Qwen, DeepSeek and Gemini to assess whether the responses produced by the 11 different sources were biased or not, and save all of these in excel files which I then used to calculate the LLM disagreement, bias rate and human-LLM disagreement.

- I analyzed the results produced by the LLMs to find useful insights, and searched for useful metrics that summarizes our findings and then used these metrics.

Amr:

- I collected prompts from the CAMEL dataset and translated them into English. I also created new prompts with Arab and general cultural contexts.
- I helped organize and analyze the human evaluation results including summarizing the final numerical findings related to cultural bias.
- Omar and I worked on writing the experimental approach, analyzing the results, and the conclusion section in the paper. We worked together to make sure that the methodology and findings were presented in a clear, structured, and easy way to follow for the reader.

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